جامعة اليمامة
Al Yamamah University

Airport Management System

CIS221: Introduction to Database System

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Phase 1— Airport Management System – Problem Definition

Airports handle many different tasks at the same time, like scheduling flights, managing passengers, handling baggage, and organizing staff. When these tasks are done manually or through separate systems that don't communicate well, it can cause delays, confusion, and more human errors. With the increasing number of flights and passengers, it becomes harder for airports to keep everything running smoothly and efficiently.

One of the main problems in airport management is the lack of integration between different departments. For example, flight scheduling systems may not be fully connected with passenger check-in or baggage handling systems. This can result in delays, lost luggage, and poor communication between airport staff. Additionally, manual processes such as recording passenger data or assigning gates can be time-consuming and prone to mistakes.

Another problem is keeping track of real-time information. Both airport staff and passengers need updated details about flight status, delays, and gate changes. Without one central system, it's hard to keep this information accurate and consistent, which can lead to confusion and a poor experience for passengers.

The Airport Management System is designed to fix this by using a centralized database that connects all airport operations in one place. This makes it easier to manage information, keep everything updated, and improve overall efficiency.

The system will manage data related to flights, passengers, staff, tickets, and baggage in a structured and efficient way. By using a relational database system, the airport can ensure data consistency, reduce redundancy, and improve overall performance.

This system will also enable better decision-making by providing accurate and real-time data. For example, airport managers can easily check flight schedules, follow passenger movement, and organize resources in a better way. Also, automating processes like check-in, ticket booking, and baggage tracking helps reduce mistakes and saves time.

Overall, the Airport Management System helps improve how the airport runs by making processes more efficient, accurate, and better connected between departments. It also provides a reliable system that can handle the increasing number of passengers while improving the experience for both staff and travelers.

Phase 2 — Database Conceptual Modeling

Main Entity Types:

- Flight (Strong entity): represents each flight in the airline system.

Primary key: idFlight and **Attributes**: departureT, ArrivalT, destination.

- Passenger (Strong entity): represents airline passengers.

Primary key: idPassenger and **Attributes**: Fname, Mint, Lname, passportNum, nationality.

- FlightCrew (Strong entity): represents airline staff/crew members.

Primary key: idStaff and **Attributes**: Fname, Mint, Lname, role, salaryMonthly.

- Tickets (Associative entity): represents a passenger booking a seat on a flight.

Primary key: idTickets, **foreign key**: flight_idFlight and **Attributes**: seat, class, price.

- Baggage (Associative entity): represents baggage belonging to passengers on flights.

Primary key: idBaggage and **foreign key**: passenger_idPassenger, flight_idFlight and **Attributes**: weight, status.

- Staff_has_flight (Associative entity): represents which crew members work on which flights.

Primary key(composite): staff_idStaff, flight_idFlight and **foreign key**: staff_idStaff, flight_idFlight.

Business Rules:

1. A passenger can book many tickets.
2. Each ticket belongs to one passenger.
3. Each ticket is linked to one flight.
4. A flight can have many tickets.
5. A passenger can have multiple baggage items.
6. Each baggage item belongs to one passenger
7. Each baggage item is assigned to one flight.
8. A flight can carry many baggage items.
9. A flight crew member can work on many flights.
10. Each flight can have multiple flight crew members.
11. Every ticket must have a unique ticket number.
12. Each flight must have a unique flight ID.

ER Diagram:

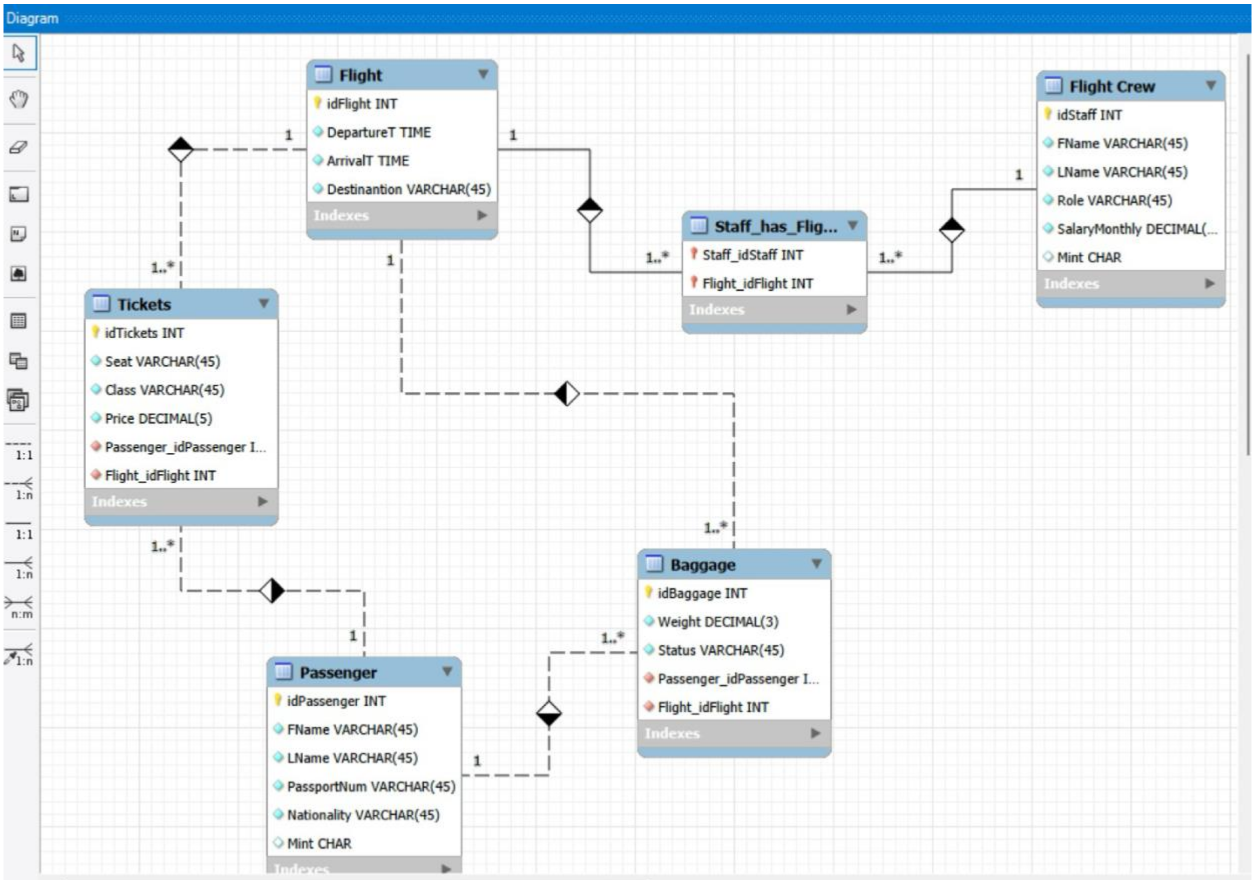
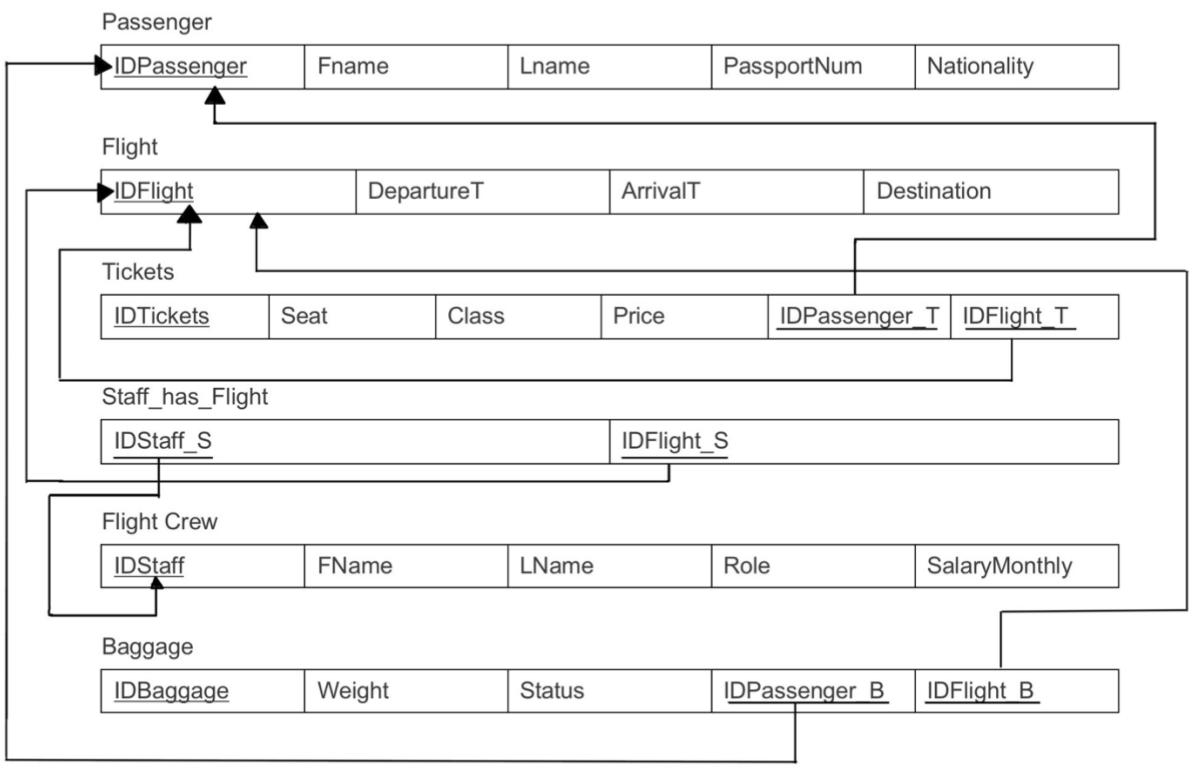


Diagram
Flight
idFlight INT
DepartureT TIME
ArrivalT TIME
Destinantion VARCHAR(45)
Indexes
Flight Crew
idStaff INT
FName VARCHAR(45)
LName VARCHAR(45)
Role VARCHAR(45)
SalaryMonthly DECIMAL(...
Mint CHAR
Indexes
Staff_has_Flig...
Staff_idStaff INT
Flight_idFlight INT
Indexes
Tickets
idTickets INT
Seat VARCHAR(45)
Class VARCHAR(45)
Price DECIMAL(5)
Passenger_idPassenger I...
Flight_idFlight INT
Indexes
Baggage
idBaggage INT
Weight DECIMAL(3)
Status VARCHAR(45)
Passenger_idPassenger I...
Flight_idFlight INT
Indexes
Passenger
idPassenger INT
FName VARCHAR(45)
LName VARCHAR(45)
PassportNum VARCHAR(45)
Nationality VARCHAR(45)
Mint CHAR
Indexes
1
1..*
1:1
1:n
n:m

Phase 3 — Database Logical Design

Mapping:



Phase 4 — Implementation

Query 1: This query displays all passenger information for those on flight en route to ‘Dubai’

SELECT *, flight_idFlight

FROM passenger, tickets, flight

WHERE flight_idFlight = idFlight and idFlight=1 and passenger_idPassenger=idPassenger;

Result:

idPassenger	Fname	Mint	Lname	passportNum	nationality	idTickets	seat	class	price	passenger_idPasseng...	flight_idFli...	idFlight	departureT	arrivalT	destination	flight_
1	Ahmed	A	Ali	P12345	Saudi	1	12A	Economy	350	1	1	1	08:00:00	10:30:00	Dubai	1
2	Sara	B	Khan	P22345	Pakistani	2	14B	Business	900	2	1	1	08:00:00	10:30:00	Dubai	1
3	John	C	Smith	P32345	American	3	10C	Economy	400	3	1	1	08:00:00	10:30:00	Dubai	1

Query 2: This query displays the total revenue of ticket sales from flight en route to ‘New York’

SELECT SUM(price) AS Total_Revenue

FROM tickets, flight

WHERE flight_idFlight = idFlight and idFlight=4;

Result:

Total_Revenue
2650

Query 3: This query displays the annual salary of all employees

SELECT fname, lname, salaryMonthly* 12 as Annual_Salary

FROM flightCrew;

Result:

fname	lname	Annual_Salary
Khalid	Salem	114000
Rana	Ali	102000
James	White	54000
Nora	Hamed	51600
Ali	Mahmoud	84000
Sophia	Clark	117600
Daniel	Lee	60000
Fatima	Saad	50400
Michael	Scott	105600
Huda	Yasin	98400

Query 4: This query displays the annual salary of all employees in descending order

select fname, lname, salaryMonthly*12 as Annual_salary

from flightCrew

ORDER BY Annual_salary DESC;

Result:

fname	lname	Annual_salary
Sophia	Clark	117600
Khalid	Salem	114000
Michael	Scott	105600
Rana	Ali	102000
Huda	Yasin	98400
Ali	Mahmoud	84000
Daniel	Lee	60000
James	White	54000
Nora	Hamed	51600
Fatima	Saad	50400

Query 5: This query displays the annual salary of all employees in ascending order

select fname, lname, salaryMonthly*12 as Annual_salary

from flightCrew

ORDER BY Annual_salary ASC;

Result:

fname	lname	Annual_salary
Fatima	Saad	50400
Nora	Hamed	51600
James	White	54000
Daniel	Lee	60000
Ali	Mahmoud	84000
Huda	Yasin	98400
Rana	Ali	102000
Michael	Scott	105600
Khalid	Salem	114000
Sophia	Clark	117600

Query 6: This displays the average weight of baggage on flight that is en route to ‘Dubai’

SELECT AVG(weight) AS Average_Baggage_Weight

FROM baggage

WHERE flight_idFlight = 1;

Result:

Average_Baggage_Weight
17.6667

Query 7: This displays the passengers whose last names contain only 5 characters

SELECT *

FROM passenger

WHERE Lname like '_____';

Result:

idPassenger	Fname	Mint	Lname	passportNum	nationality
3	John	C	Smith	P32345	American
7	Adam	G	Brown	P72345	British
8	Noor	H	Saleh	P82345	Saudi
NULL	NULL	NULL	NULL	NULL	NULL

Query 8: Displays the tickets that have average prices higher than 500 in descending order

SELECT class, AVG(price) AS Average_Price

FROM tickets

GROUP BY class

HAVING AVG(price) > 500

ORDER BY Average_Price DESC;

Result:

class	Average_Price
First	1600.0000
Business	900.0000

Query 9: Displays the lowest weighing baggage on flight en route to ‘Dubai’

SELECT weight, passenger_idPassenger, flight_idFlight

FROM baggage

WHERE flight_idFlight = 1;

(

SELECT MIN(weight)

FROM baggage

WHERE flight_idFlight = 1

);

Result:

MIN(weight)
15

Query 10: Displays the status of all passenger baggage, and which flight they were on

```
SELECT status, Fname, Lname, idFlight, destination, departureT
FROM baggage
JOIN passenger
   ON passenger_idPassenger = idPassenger
JOIN flight
   ON flight_idFlight = idFlight;
```

Result:

status	Fname	Lname	idFlight	destination	departureT
Checked	Ahmed	Ali	1	Dubai	08:00:00
Loaded	Sara	Khan	1	Dubai	08:00:00
Checked	John	Smith	1	Dubai	08:00:00
Delayed	Lina	Hassan	2	London	09:15:00
Loaded	Omar	Farouk	2	London	09:15:00
Checked	Maya	George	3	Paris	11:45:00
Lost	Adam	Brown	3	Paris	11:45:00
Loaded	Noor	Saleh	4	New York	13:00:00
Checked	Emma	Wilson	4	New York	13:00:00
Loaded	Yousef	Nasser	5	Cairo	15:30:00

Query 11: Displays the most expensive ticket purchased on flight en route to ‘Paris’

```sql
SELECT MAX(Price) AS Max_Price
FROM tickets
WHERE flight_idFlight = 3;
```

Result:

Max_Price
850